PHYSICS 2VG3 - Homework 3 Report

Exercise 1 - Off-centre collision between two equal masses
Setup: m1=m2=1.0, r1=r2=1.0
Initial positions: (-5,25,0.4) and (5,25,-0.4)
Initial velocities: v1=(2,0,0), v2=(-1,0,0)

Qualitative outcome:
The two equal masses exchange their normal velocity components at contact.
The left mass deflects upward in +z and moves left; the right mass deflects downward in -z and moves right.

Numerical results from collision_2mass.py:
Before collision:
  P = (1, 0, 0)
  L = (0, 1.2, -25)
  K = 2.5
Immediately after first collision:
  P = (1, 0, 0)
  L = (0, 1.2002916688, -25)
  K = 2.4999999999999987
After simulation:
  P = (1, 0, 0)
  L = (0, 1.2002916688, -25)
  K = 2.4999999999999987
Final velocities:
  v_left = (-0.5197666366, 0, 1.10003454804)
  v_right = (1.5197666366, 0, -1.10003454804)

Exercise 2 - Basketball, ping-pong ball, wall
Setup: inverse masses = 1.0, 0.01, 0.0 (wall)
Why inverse mass of wall is 0: infinite mass means zero acceleration under finite impulse,
so the wall remains fixed (physically appropriate rigid boundary).

Given positions/radii/velocities from assignment:
Final velocities from collision_3masses.py baseline:
  v1 (ping-pong) = (-29.603960396, 0, 0)
  v2 (basketball) = (-9.60396039604, 0, 0)
  v3 (wall)   = (0, 0, 0)
As multiples of initial vx=10:
  v1/vx = -2.9603960396039604
  v2/vx = -0.9603960396039604
  v3/vx = 0

Sequence of events:
1) The basketball (mass 100) hits the wall and reverses direction.
2) It then collides head-on with the ping-pong ball (mass 1).
3) Elastic collision transfers large speed change to the light ball.

Minor setup changes (no overlap, same order, same initial speed) affect timing only;
the normalized final speeds v1/vx and v2/vx remain the same in these tests.

General symbolic result for any initial vx=u>0 with masses m (left) and M (middle):
v1 = ((m - 3M)/(m + M)) u
v2 = ((3m - M)/(m + M)) u
v3 = 0
Limiting case m << M (very light first mass):
v1 -> -3u, v2 -> -u, v3 = 0

Exercise 3 - Many random particles in a 3D box
Code file: collsion_many.py (12 particles, random seed=2)
All simulations now run as visual Panda3D apps (ShowBase + task loop).
Walls used at x,z in [-20,20], y in [80,120], with elastic bounce in all 3 directions.

Conserved quantities printed by code:
Start (t=0):
P = (7.21969973911, 39.6792874611, 1.8345113423)
L = (287.697232866, 168.487732251, -195.501279954)
K = 669.8171214683324
After 30 s:
P = (-23.9514357434, -25.062591668, -35.3678347096)
L = (-3989.53484698, 221.370669718, 2735.20412379)
K = 669.8171214683325

What is conserved now:
Kinetic energy is conserved (elastic particle-particle and particle-wall collisions).
Linear momentum is not conserved for the particle subsystem because walls exert external impulses.
Angular momentum is also not conserved for the subsystem because wall impulses provide external torque.

Submitted code files:
panda_collision.py
collision_2masses.py
collision_2mass.py
collision_3masses.py
collsion_many.py